

CO4 - AT3 - COMPARATIVE ANALYSIS

Course Code: ITA0609

Course Title: Machine Learning

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1). Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

Sol:- Introduction

Case-Based Learning (CBL) and Locally Weighted Regression (LWR) are machine learning approaches that use previous data to make predictions for new situations. Both methods are useful in recommendation systems because they can use information from similar users or items. However, they differ in how they process new data, adapt to changes, and handle computational requirements.

Practical Example: Movie Recommendation System

Consider a movie recommendation system that recommends movies to users based on their previous ratings and preferences.

For example, if a user likes action and science-fiction movies, the system can use information from similar users to recommend movies that the user may like.

1. Case-Based Learning

Case-Based Learning stores previous experiences or cases and solves a new problem by finding similar previous cases.

Example

Suppose a new user has rated the following movies:

- Interstellar – 5 stars
- Avengers – 5 stars
- Inception – 4 stars

The system searches its database for users with similar movie preferences. If a similar user has given a high rating to "The Matrix", the system can recommend "The Matrix" to the new user.

Working

1. Store previous user preferences and ratings.
2. Receive the new user's preferences.
3. Search for similar users or cases.

4. Identify recommendations from similar cases.
5. Recommend suitable movies.
6. Store the new user's case for future recommendations.

2. Locally Weighted Regression

Locally Weighted Regression predicts an output by giving more importance to data points that are close or similar to the new input.

In a recommendation system, users or movies that are more similar to the current user are given higher weights.

Example

Suppose a new user has similar preferences to three existing users. The system assigns higher weights to the most similar users and lower weights to less similar users. It then predicts the rating of an unseen movie using these weighted values.

For example:

- User A similarity = 0.9 → High weight
- User B similarity = 0.7 → Medium weight
- User C similarity = 0.2 → Low weight

The prediction is mainly influenced by User A because the user is more similar to the new user.

Comparison of Case-Based Learning and Locally Weighted Regression

Feature	Case-Based Learning	Locally Weighted Regression
Basic concept	Finds and reuses similar previous cases	Uses nearby data points to make a prediction
Working principle	Retrieval and reuse	Weighted local regression
New data handling	New cases can be added directly to the case database	New data can influence local predictions without complete retraining
Adaptability	Very high	Very high
Training requirement	Very little training is required	Little or no global training
Prediction	Based on similar previous cases	Based on weighted nearby data
Computational efficiency	Can be slow when searching a large number of cases	Can be computationally expensive for large datasets
Memory requirement	Requires storage of previous cases	Usually requires storage of training data

Best suited for	Problems where previous solutions can be reused	Problems involving continuous numerical prediction
Example	Recommend movies based on similar users	Predict a user's rating for a movie
Handling changes	Easily adapts by adding new cases	Adapts through updated local data
Complexity	Mainly depends on case retrieval and similarity calculation	Depends on the number of data points and local regression calculation

Handling of New Data

Case-Based Learning

CBL handles new data very easily. When a new user or case becomes available, it can simply be added to the case database. The system does not need to retrain a complete model.

Locally Weighted Regression

LWR also handles new data well because predictions are based on nearby training examples. New data can be included in future predictions without rebuilding a global model.

Adaptability

Case-Based Learning is highly adaptable because it can continuously store new experiences. As more users interact with the recommendation system, new cases can be added and used for future recommendations.

Locally Weighted Regression is also highly adaptable because it focuses on local patterns. It can respond to changes in user preferences, especially when recent or nearby data is given greater importance.

Computational Efficiency

Case-Based Learning can become computationally expensive when the database contains a very large number of cases because the system needs to search for similar cases.

Locally Weighted Regression can also require significant computation because distances and weights may need to be calculated for many training points for every new prediction.

Therefore, both methods are suitable for small and medium-sized datasets, but optimization techniques such as indexing, selecting important data points, or reducing the dataset may be required for very large datasets.

Advantages

Case-Based Learning

- Simple and easy to understand.
- Easily incorporates new cases.
- Does not require extensive model training.

- Can provide recommendations based on actual previous cases.
- Useful when previous solutions can be reused.

Locally Weighted Regression

- Handles nonlinear relationships effectively.
- Gives greater importance to similar data points.
- Adapts well to changing data.
- Useful for predicting numerical ratings.
- Does not require a single global model.

Conclusion

Case-Based Learning and Locally Weighted Regression are both useful approaches for recommendation systems. Case-Based Learning retrieves similar previous cases and reuses their solutions, while Locally Weighted Regression assigns weights to nearby data points and uses them to predict an output. Both approaches can handle new data and adapt to changing user preferences. However, their computational cost can increase as the dataset becomes larger. CBL is more suitable when previous cases can be directly reused, whereas LWR is more suitable when a numerical prediction, such as a movie rating, needs to be calculated.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

Sol:-Introduction

Locally Weighted Regression (LWR) and Radial Basis Function (RBF) Networks are machine learning techniques used for function approximation. Both methods can model nonlinear relationships, but they differ in locality, computational cost, training process, and smoothness of predictions.

Example Dataset

Consider the following dataset:

X	Y
0	0.00
1	0.84
2	0.91
3	0.14
4	-0.76
5	-0.96
6	-0.28
7	0.66
8	0.99
9	0.41

The objective is to learn the relationship between X and Y and predict Y for a new value such as $X = 4.5$.

Locally Weighted Regression

Locally Weighted Regression is a non-parametric method that gives more importance to data points that are close to the query point.

For example, when predicting the output for $X = 4.5$, the points $X = 4$ and $X = 5$ receive higher weights because they are closer to 4.5. Points such as $X = 0$ and $X = 9$ receive lower weights.

The weight can be calculated using:

$$w_i = \exp(-((x_i - x_q)^2)/(2\tau^2))$$

Where:

- x_i = training data point
- x_q = query point
- τ = bandwidth parameter
- w_i = weight of the data point

LWR performs a local regression using these weighted data points.

Radial Basis Function Network

A Radial Basis Function Network is a neural network commonly used for nonlinear function approximation. It consists of an input layer, hidden layer, and output layer.

The hidden layer uses radial basis functions, commonly Gaussian functions.

The Gaussian function is:

$$\phi(x) = \exp(-((x-c)^2)/(2\sigma^2))$$

Where:

- x = input
- c = center of the RBF
- σ = width of the RBF
- $\phi(x)$ = activation value

For $X = 4.5$, an RBF with a center close to 4.5 produces a strong output, while RBFs with centers farther away produce weaker outputs. The outputs are combined using learned weights to produce the final prediction.

Comparison

Feature	Locally Weighted Regression	Radial Basis Function Network
Basic principle	Uses nearby data points for prediction	Uses radial basis functions
Locality	Dynamically calculated for every query	Determined by RBF centers and widths
Training	Little or no global training	Requires training
Prediction	Performs local regression for each query	Uses the trained network
Computational cost	High during prediction for large datasets	Higher during training, but faster prediction
Adaptability	Very high	Moderate
Memory	Stores training data	Stores centers, widths, and weights
Nonlinear approximation	Good	Very good
Smoothness	Depends on bandwidth	Generally smooth
Large datasets	Can become slow	More efficient after training
Main parameters	Bandwidth (τ)	Centers, width (σ), and weights

Difference in Locality

In LWR, locality is determined dynamically. For every new input, the distance between the input and training points is calculated and weights are assigned.

In an RBF Network, locality is determined by the centers and widths of the radial basis functions. Each RBF responds strongly to inputs near its center.

Therefore, **LWR has dynamic locality, whereas RBF Networks have learned or predefined locality.**

Difference in Computational Cost

LWR requires distance and weight calculations for training points whenever a prediction is made. Therefore, prediction becomes computationally expensive when the dataset is large.

RBF Networks perform most of their computation during training. Once trained, they can make predictions quickly using the learned parameters.

Therefore:

LWR → Low training cost but higher prediction cost

RBF Network → Higher training cost but lower prediction cost

Difference in Smoothness

The smoothness of LWR depends on the bandwidth parameter.

- Small bandwidth → More local and less smooth predictions.
- Large bandwidth → Smoother predictions but less local behavior.

RBF Networks generally provide smooth predictions because Gaussian functions change gradually with the input.

- Small width → More localized prediction.
- Large width → Smoother prediction.

Conclusion

Locally Weighted Regression and Radial Basis Function Networks are both effective for function approximation. LWR is highly adaptable and focuses directly on nearby training data, making it useful for small or changing datasets. However, it can become computationally expensive during prediction. RBF Networks require training but provide smooth predictions and faster computation after training. Hence, LWR is preferred when adaptability and local fitting are important, while RBF Networks are preferred when smooth function approximation and fast repeated predictions are required.

